

```
%root_path = fileparts(mfilename("fullpath"));
%root_path = pwd;

% retrieve selected cuts used for further fitting
[GH_FFCutsFits,CutFitFileGH] =
recover_selected_cuts('GH_FFCutsFits','e:\SHARE\FE\Data\analysis\07_fit_with_multicu
t_all\sel_cuts');
```

```
fn = fieldnames(GH_FFCutsFits);
%nf = 11
%for nf = 2:numel(fn)
%selectCuts = fn{nf};
selectCuts = 'Ei400_off110GHmaxFF';
if ismember(selectCuts(1:3),{'msk','dat'})||...
    contains(selectCuts,'all_fit_par')
    %continue
    error('Can not fit field: %s',selectCuts);
end

fprintf('*****\n');
```

```
*****
```

```
fprintf('Processing cuts: %s \n', selectCuts);
```

```
Processing cuts: Ei400_off110GHminFF
```

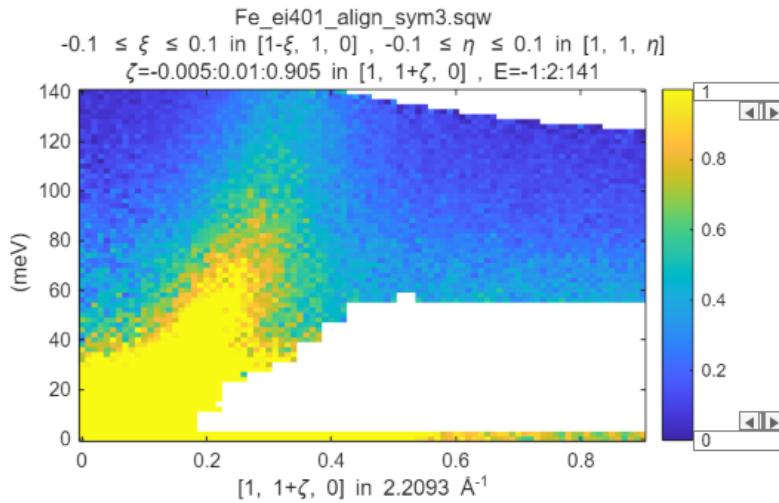
```
fprintf('*****\n');
```

```
*****
```

```
cuts_list = GH_FFCutsFits.(selectCuts);
if ~iscell(cuts_list)
    cuts_list = {cuts_list};
end
if contains(selectCuts,'Ei200')
    min_Ei = 30;
    mi = maps_instrument(200,600,'S');
end
if contains(selectCuts,'Ei400')
    min_Ei = 40;
    mi = maps_instrument(401,600,'S');
end
if contains(selectCuts,'Ei800')
    min_Ei = 70;
    mi = maps_instrument(786,600,'S');
end
sample=IX_sample(true,[1,0,0],[0,1,0],'cuboid',[0.04,0.03,0.02]);
for j=1:numel(cuts_list)
```

```
cuts_list{j} = cuts_list{j}.set_instrument(mi);
cuts_list{j} = cuts_list{j}.set_sample(sample);
plot(cuts_list{j}); lz 0 1; keep_figure;
```

end



```
dir_name = "GH";

dE_step = 2; %original energy transfer step data were binned to. No point in going
finer
half_dE = 5; % half width of data binning

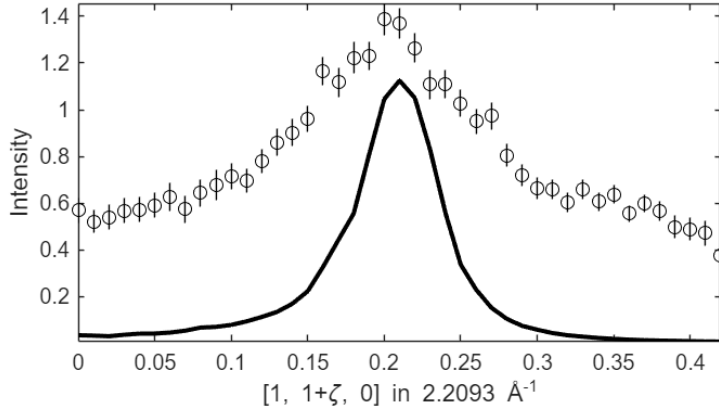
en_mm = cuts_list{1}.data.img_range(:,4);
en_mm(1) = max(min_Ei,en_mm(1));
%en_mm(1) = 100;
%en_mm(2) = 170;
cut_en = en_mm(1)+half_dE:2*half_dE:(en_mm(2)-half_dE);

cut_dimensions = 1;
GH_FFCutsFits.(selectCuts) = cuts_list;

GH_FFCutsFits = fit_multicutsFF_along_direction(...
    GH_FFCutsFits,selectCuts,cut_dimensions,cut_en,dE_step,half_dE,true);
```

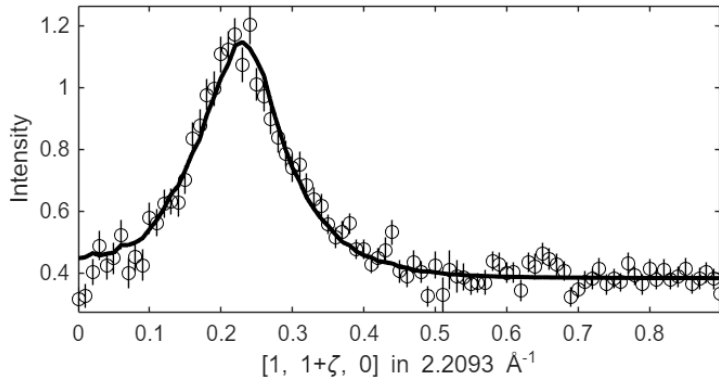
```
*****
**** En = 45±5 going to 135
*****
```

Fe_ei401_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $40 \leq E \leq 135$ eV
 $\zeta = -0.005:0.01:0.905$ in $[1, 1+\zeta, 0]$

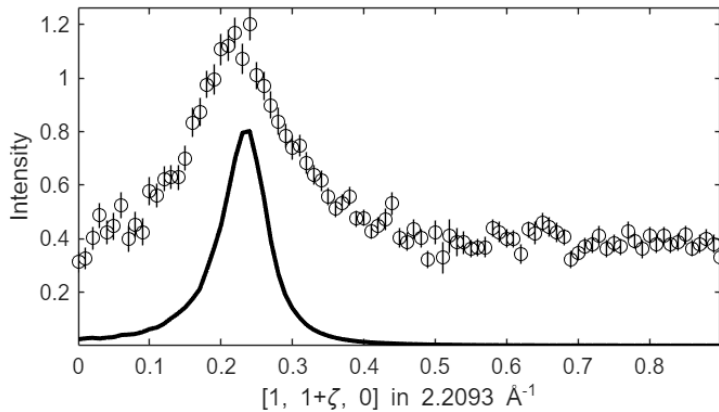


 **** En = 55±5 going to 135

Fe_ei401_align_sym3.sqw
 $S = 0.8 \pm 0.018$; $J_0 = 25 \pm 0.34$; $J_1 = 0 \pm 0$; $\gamma = 12.9474 \pm 0.325788$
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $50 \leq E \leq 135$ eV
 $\zeta = -0.005:0.01:0.905$ in $[1, 1+\zeta, 0]$



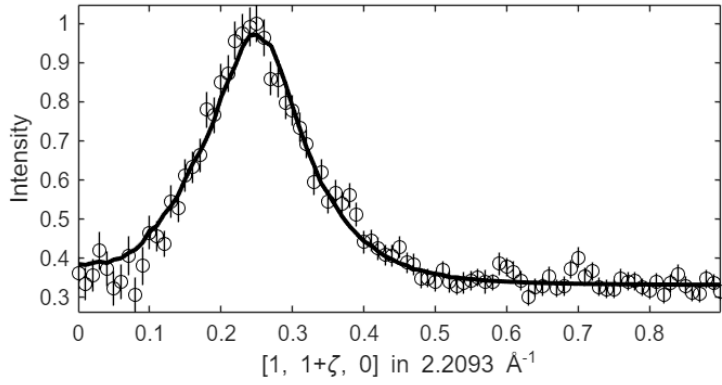
Fe_ei401_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $50 \leq E \leq 135$ eV
 $\zeta = -0.005:0.01:0.905$ in $[1, 1+\zeta, 0]$



 **** En = 65±5 going to 135

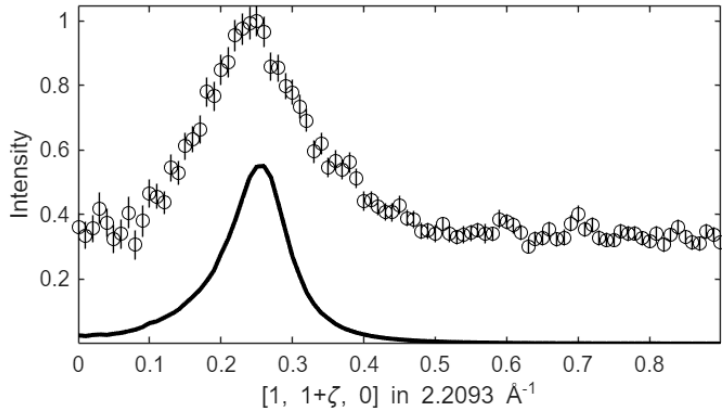
Fe_ei401_align_sym3.sqw

S= 0.9±0.015; J0= 26±0.27; J1= 0± 0; gamma=21.8195±0.391867
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $60 \leq E \leq 70$ eV
 $\zeta=-0.005:0.01:0.905$ in $[1, 1+\zeta, 0]$



Fe_ei401_align_sym3.sqw

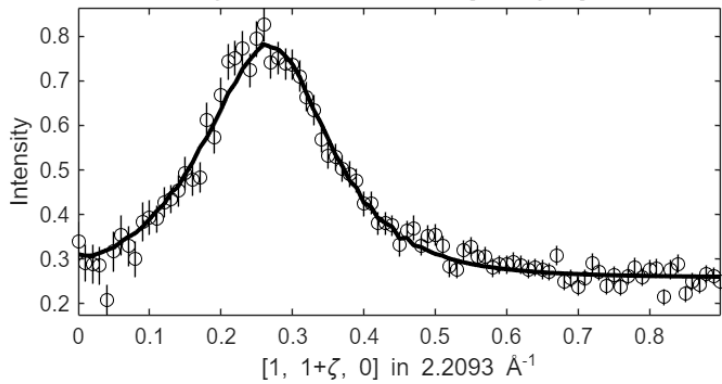
$-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $60 \leq E \leq 70$ eV
 $\zeta=-0.005:0.01:0.905$ in $[1, 1+\zeta, 0]$



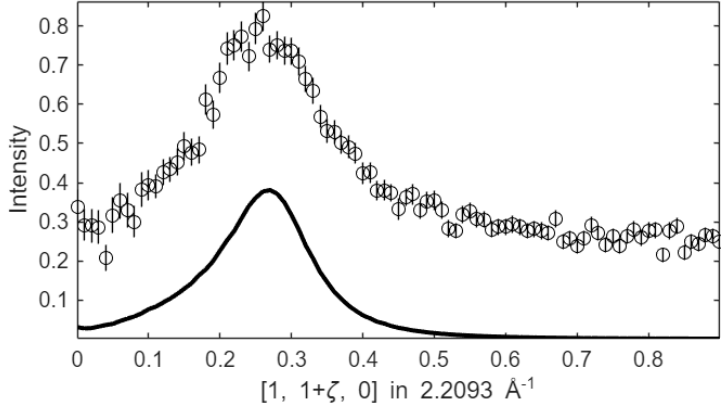
**** En = 75±5 going to 135

Fe_ei401_align_sym3.sqw

S= 1.2±0.093; J0= 30±0.96; J1= 0± 0; gamma=46.3473±5.44613
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $70 \leq E \leq 80$ eV
 $\zeta=-0.005:0.01:0.905$ in $[1, 1+\zeta, 0]$

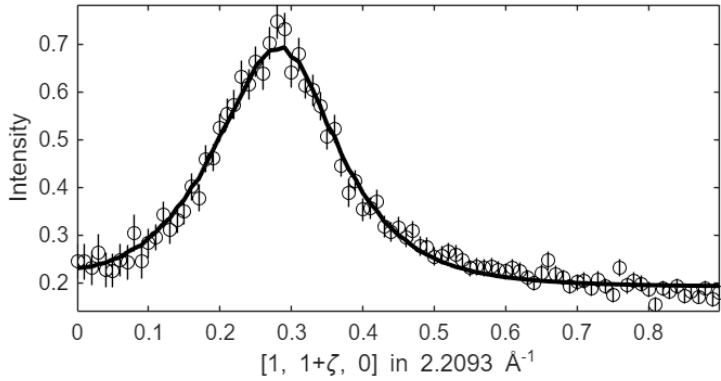


Fe_ei401_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $70 \leq E \leq 100$ eV
 $\zeta = -0.005:0.01:0.905$ in $[1, 1+\zeta, 0]$

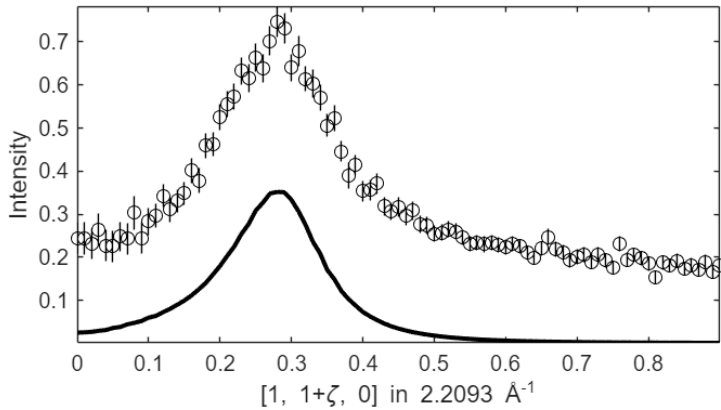


 **** En = 85±5 going to 135

Fe_ei401_align_sym3.sqw
 $S = 1.2 \pm 0.018$; $J_0 = 30 \pm 0.23$; $J_1 = 0 \pm 0$; $\gamma = 48.7445 \pm 0.626028$
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $80 \leq E \leq 100$ eV
 $\zeta = -0.005:0.01:0.905$ in $[1, 1+\zeta, 0]$



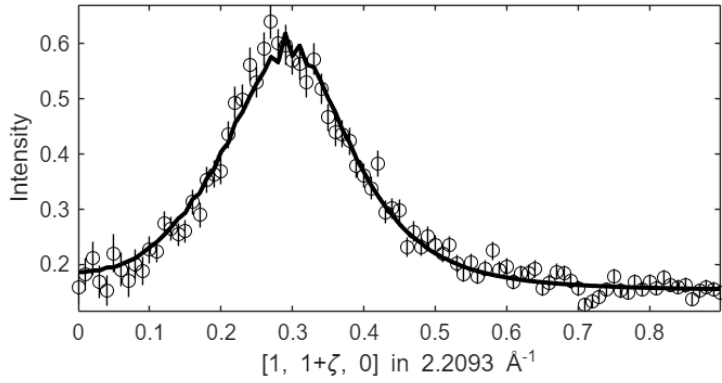
Fe_ei401_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $80 \leq E \leq 100$ eV
 $\zeta = -0.005:0.01:0.905$ in $[1, 1+\zeta, 0]$



 **** En = 95±5 going to 135

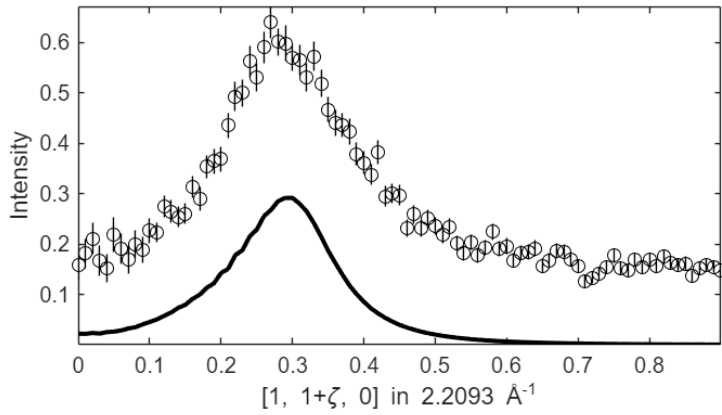
Fe_ei401_align_sym3.sqw

S= 1.2±0.02; J0= 32±0.27; J1= 0± 0; gamma=57.9431±0.865025
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $90 \leq E \leq 1$
 $\zeta=-0.005:0.01:0.905$ in $[1, 1+\zeta, 0]$



Fe_ei401_align_sym3.sqw

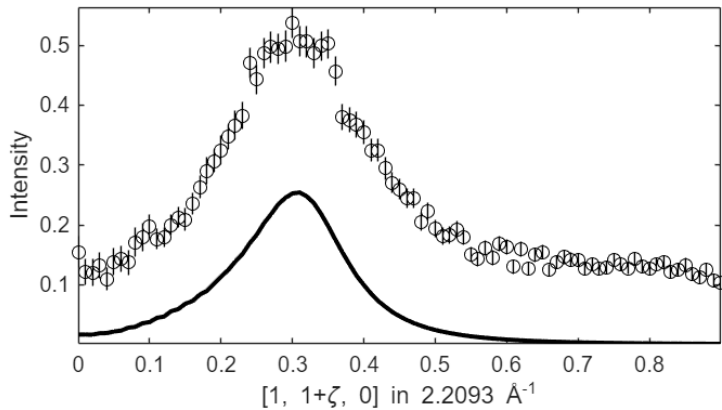
$-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $90 \leq E \leq 1$
 $\zeta=-0.005:0.01:0.905$ in $[1, 1+\zeta, 0]$



 **** En = 105±5 going to 135

Fe_ei401_align_sym3.sqw

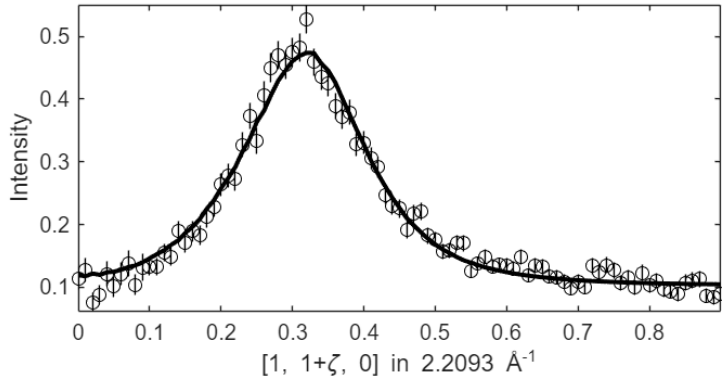
$0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $100 \leq E \leq$
 $\zeta=-0.005:0.01:0.905$ in $[1, 1+\zeta, 0]$



 **** En = 115±5 going to 135

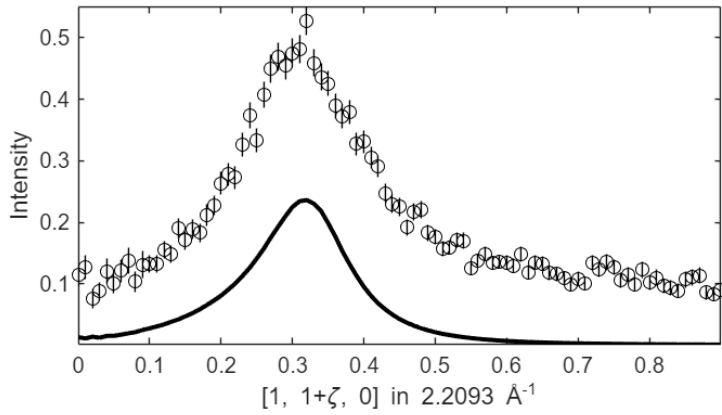
Fe_ei401_align_sym3.sqw

S= 1 ± 0.019 ; J0= 32 ± 0.24 ; J1= 0 ± 0 ; gamma= 55.8823 ± 0.959044
 $0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $110 \leq E \leq$
 $\zeta = -0.005:0.01:0.905$ in $[1, 1+\zeta, 0]$



Fe_ei401_align_sym3.sqw

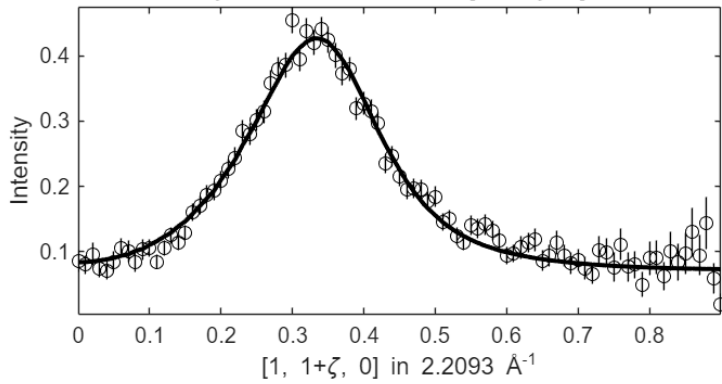
$0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $110 \leq E \leq$
 $\zeta = -0.005:0.01:0.905$ in $[1, 1+\zeta, 0]$



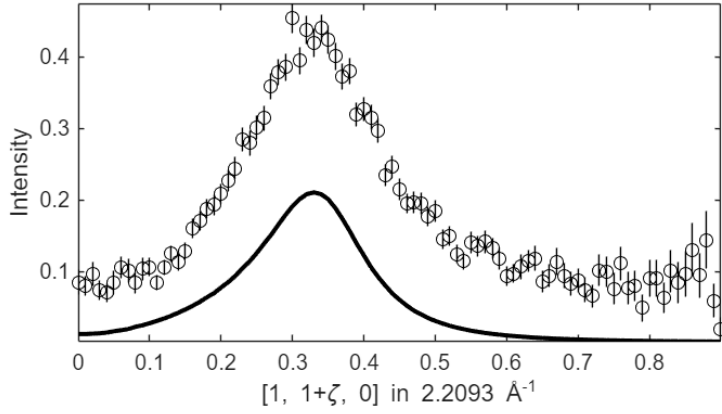
**** En = 125 ± 5 going to 135

Fe_ei401_align_sym3.sqw

S= 1.2 ± 0.033 ; J0= 34 ± 0.26 ; J1= 0 ± 0 ; gamma= 73.4613 ± 1.80914
 $0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $120 \leq E \leq$
 $\zeta = -0.005:0.01:0.905$ in $[1, 1+\zeta, 0]$



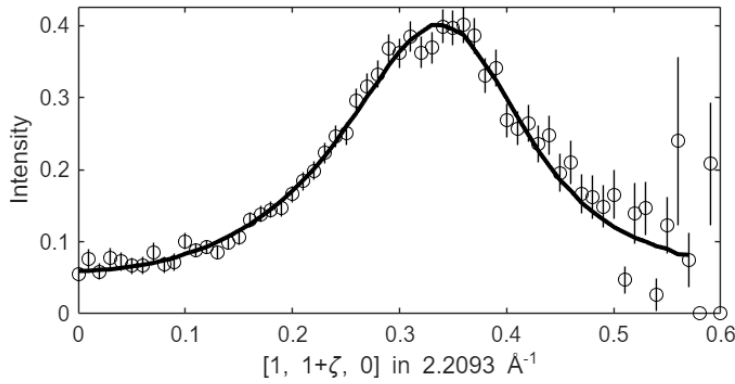
Fe_ei401_align_sym3.sqw
 $0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $120 \leq E \leq$
 $\zeta = -0.005:0.01:0.905$ in $[1, 1+\zeta, 0]$



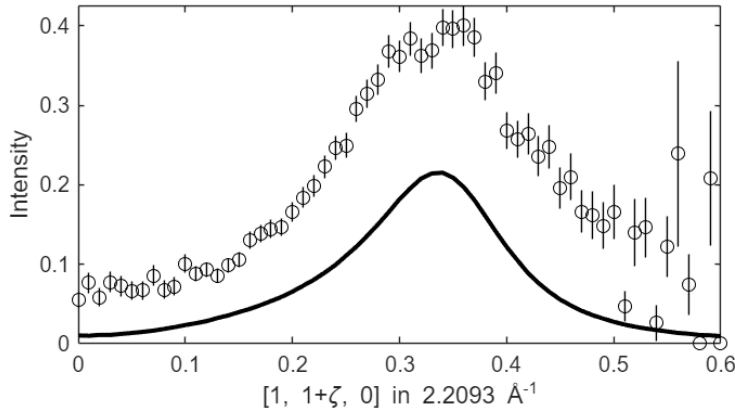
**** En = 135±5 going to 135

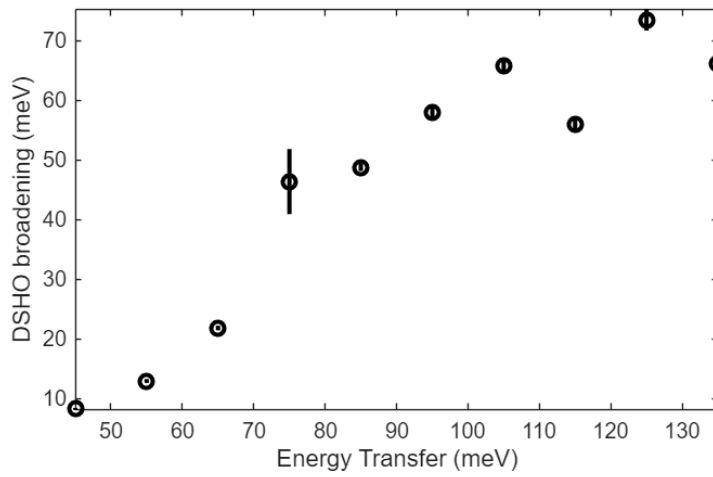
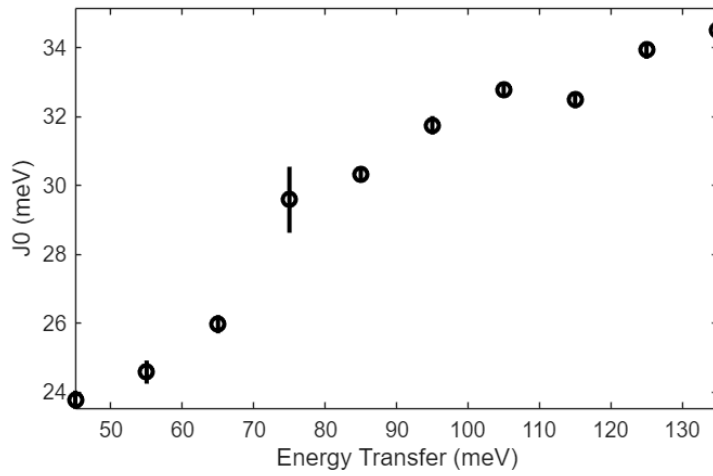
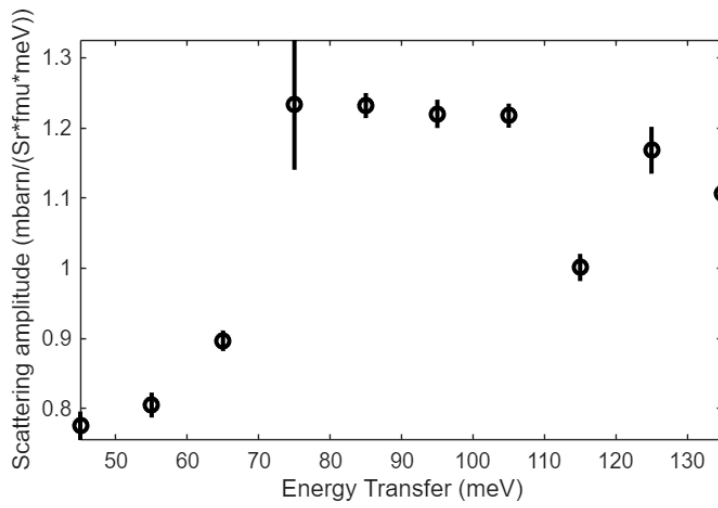
Warning: Dataset:2 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 2.1978

Fe_ei401_align_sym3.sqw
 $S = 1.1 \pm 0.079$; $J_0 = 35 \pm 0.64$; $J_1 = 0 \pm 0$; $\gamma = 66.0295 \pm 6.44205$
 $0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $130 \leq E \leq$
 $\zeta = -0.005:0.01:0.905$ in $[1, 1+\zeta, 0]$



Fe_ei401_align_sym3.sqw
 $0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $130 \leq E \leq$
 $\zeta = -0.005:0.01:0.905$ in $[1, 1+\zeta, 0]$





```

c1 = GH_FFCutsFits.(selectCuts);
GH_FFCutsFits.(selectCuts) = [c1{:}];

beep
beep
beep

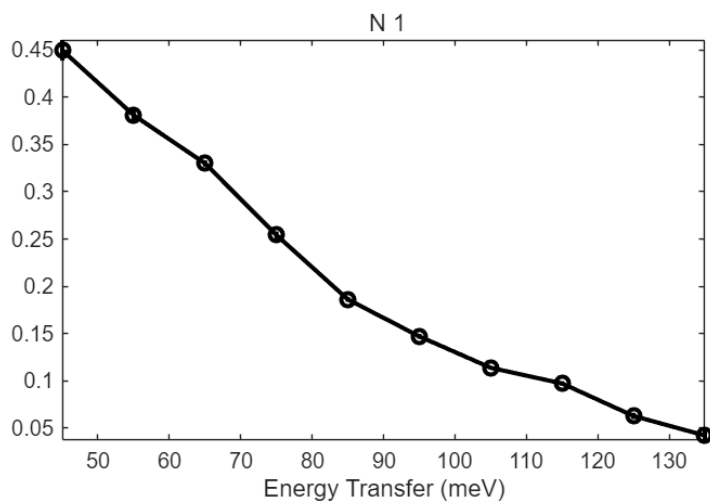
all_fit_par = GH_FFCutsFits.([selectCuts, '_all_fit_par']);

```

```

bg_data = extract_bg_par(all_fit_par,[1,2]);
% original slope parameters
pd(bg_data(1));lx(all_fit_par(1).en,all_fit_par(end).en);keep_figure;

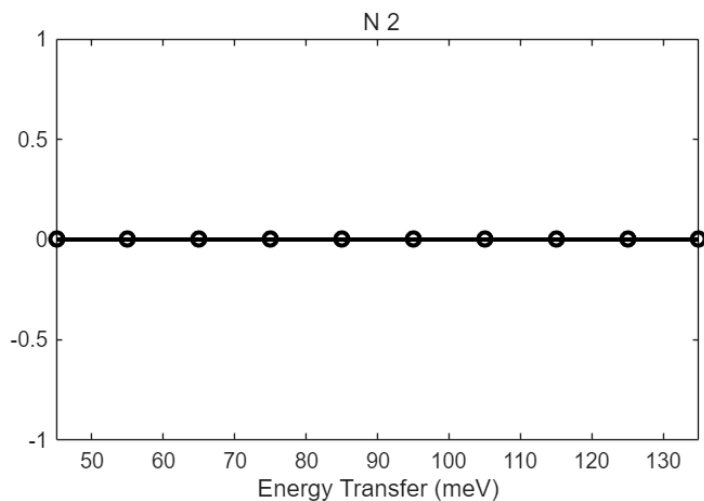
```



```

pd(bg_data(2));lx(all_fit_par(1).en,all_fit_par(end).en);keep_figure

```



```

%pd(bg_data(3));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_fig
ure;
%pd(bg_data(4));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_fig
ure

```

```

doc = matlab.desktop.editor.getActive;
doc.save;
thisFile = 'e:\SHARE\Fe\Data\analysis\06_fit_with_AQ0\fit_cutFe_GH.m';
targ_file = fullfile(fileparts(thisFile),['fit_cutFe_GH_',selectCuts,'.pdf']);
export(thisFile,targ_file);

```

```

fit_src_struct = GH_FFCutsFits;
save(CutFitFileGH,"fit_src_struct",-v7.3');

```

```
close all;  
%end
```